# UNIVERSITY OF KELANIYA - SRI LANKA
# FACULTY OF SCIENCE

**Bachelor of Science Honours Degree Examination, July/August - 2025**

Academic Year: 2023/2024 - Semester I

**ELECTRONICS AND COMPUTER SCIENCE**

**BECS 21413/BECS 21413 (R) - Analogue Electronics II (Operational Amplifiers)**

**No. of Questions: Seven (07)** **No. of Pages: Five (05)** **Time: 02 hours & 30 minutes**

Answer any **FIVE (05)** questions.

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***(All the symbols and abbreviations have their usual meanings)***

1) (a) Describe the op-amp's negative feedback concept using a suitable diagram. (*04 Marks*)

(b) Using circuit diagrams, derive expressions for the output voltage ($V_o$) of op-amp's basic inverting and non-inverting closed-loop configurations. Using the derived expressions, obtain closed-loop gains for inverting and non-inverting configurations. (*08 Marks*)

(c) Using the circuit diagrams and the expressions you have obtained in part (b), discuss the construction of inverter and voltage follower circuits. (*04 Marks*)

(d) Figure 1 shows a circuit diagram of an AC inverting amplifier with a single power supply.

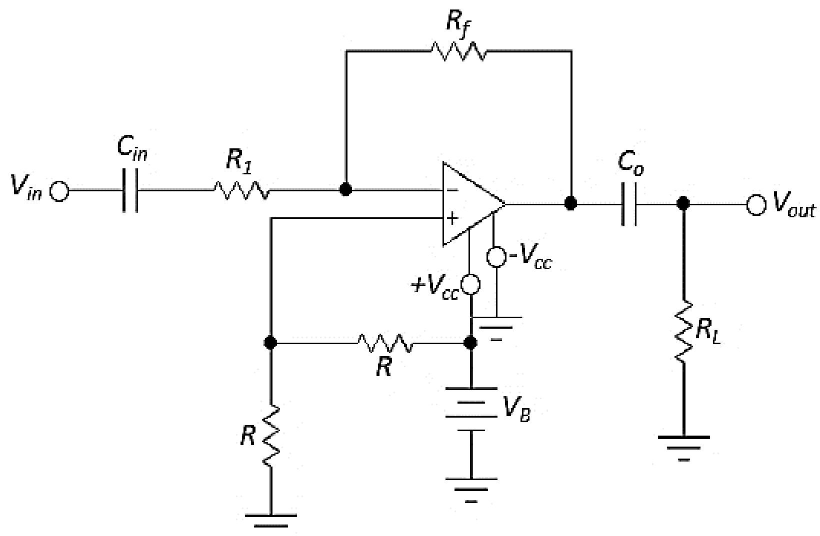


*Figure 1*

(i) Draw the DC equivalent circuit of the amplifier and show that the output AC signal swings about the DC level of ${V_B}/{2}$ . (*05 Marks*)

(ii) Draw the AC equivalent circuit of the amplifier and show that only the AC signal appears across the load $R_L$. (*04 Marks*)

2) (a) The circuit diagram of a true integrator is shown in Figure 2. Show that the output is the integration of the input function by deriving an expression for the output voltage of the circuit. (*05 Marks*)

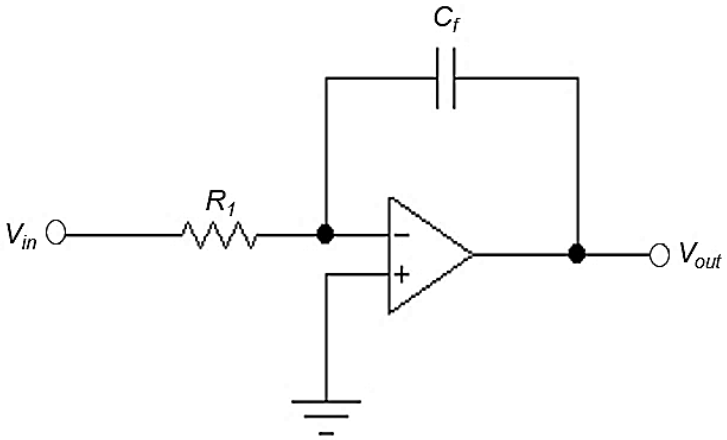


*Figure 2*

(b) Sketch the output function of the true integrator if a constant DC voltage of 1 V is applied to the input at $t = 0$. Hence, find an expression for the gradient of the plot. (*04 Marks*)

(c) Derive the transfer function of the true integrator and discuss its main design issue in response to a very low-frequency input signals. (*06 Marks*)

(d) Modify the above true integrator to a practical integrator circuit to mitigate the above design issue and derive its transfer function. (*07 Marks*)

(e) Hence, show that the practical integrator has a low-pass frequency response. (*03 Marks*)

3) The circuit diagram of a phase shifter is shown in Figure 3.

(a) By applying the superposition theorem, show that the transfer function of the phase shifter is given by,

$$T(jf) = \frac{1 - j\left({f}/{f_c}\right)}{1 + j\left({f}/{f_c}\right)} \quad ; \quad f_c = \frac{1}{2\pi RC} \ .$$

(*10 Marks*)

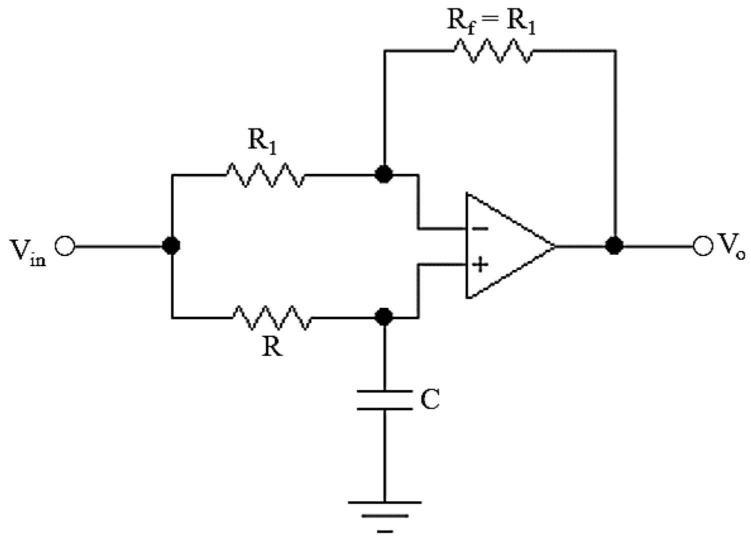


*Figure 3*

(b) Determine the amplitude response of the phase shifter and, hence, explain how the phase shifter becomes an all-pass filter. (*05 Marks*)

(c) Determine the phase response and state the type of the phase shifter. (*05 Marks*)

(d) If $R_1 = 10\ k\Omega$, $R = 2\ k\Omega$ and $C = 0.01\ \mu F$, calculate the frequency of the input signal when the phase shift between input and output becomes $90^o$. (*05 Marks*)

4) (a) Draw the first-order RC passive low-pass and high-pass filter circuits and derive their transfer functions. Hence, obtain expressions for the cut-off frequency of both filters (*06 Marks*)

(b) Modify the above two circuits using an op-amp to have active gain control (non-inverting) and obtain the transfer functions for new circuits. (*06 Marks*)

(c) Using the results of parts (a) and (b), design an active (non-inverting) band-pass filter with a ***single-stage gain control*** and obtain its transfer function. (*08 Marks*)

(d) Design an active first-order band-pass filter having a pass band of 1 to 10 kHz and a gain of 2. (*05 Marks*)

5) (a) Describe the inverting and non-inverting modes of an analogue voltage comparator. (*06 Marks*)

(b) Using suitable diagrams, describe how an analogue voltage comparator becomes zero-level and voltage-threshold detectors. (*06 Marks*)

(c) Soil moisture is a measure of the water content of the soil. Since the soil solution is the medium for the supply of nutrients to the plants, soil moisture is a critical parameter for the plants' growth. In a given application, there are four plant growth fields, each with a water spraying system and soil moisture measuring sensor. Suppose the soil moisture level of a field is below the average soil moisture level of all four fields. In that case, the water spraying system of that field must be automatically turned *ON* and *OFF* when soil moisture goes up to the required level (average soil moisture level of all four fields).

(i) Design an op-amp-based circuit to measure the average of the sensor outputs of all four soil moisture sensors. Note that each sensor produces a positive voltage proportional to the soil moisture content of the respective field. (*07 Marks*)

(ii) Design a circuit to control the soil moisture level of all four fields. Sensor outputs are the inputs of the circuit and outputs of the circuit switch *ON* and *OFF* the water spray system in each field. (*06 Marks*)

6) Figure 4 shows the circuit diagram of a Wien-bridge network.

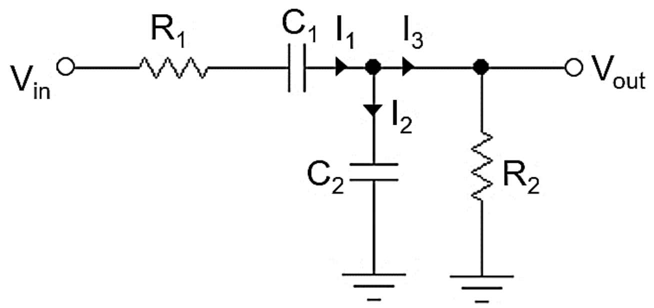


*Figure 4*

(a) Derive the transfer function of the Wien-bridge network shown in Figure 4 and hence, obtain an expression for the resonant frequency when $R_1 = R_2 = R$ and $C_1 = C_2 = C$. (*10 Marks*)

(b) Draw the Wien-bridge oscillator circuit using an op-amp and describe the conditions for oscillation. (*08 Marks*)

(c) Design a Wien-bridge oscillator circuit having an oscillation frequency of 1 kHz. Use realistic values for all passive components. (*07 Marks*)

7) (a) Define the terms "Line regulation" and Load regulation" in relation to the voltage regulation. (*04 Marks*)

(b) Determine the output voltage of the regulator shown in Figure 5. (*04 Marks*)

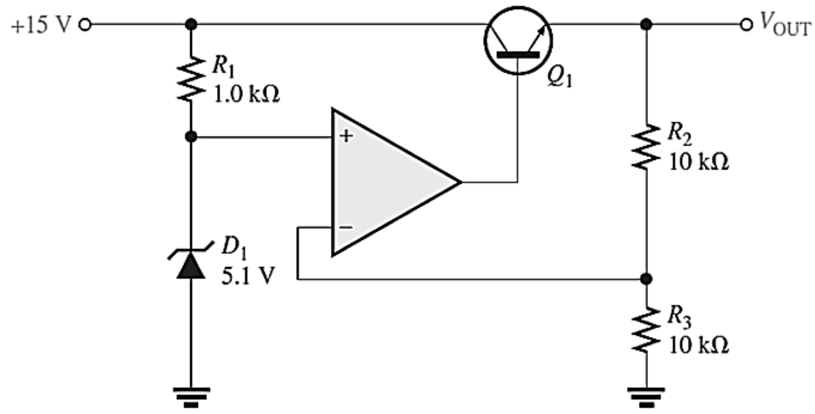


*Figure 5*

(c) Modify the regulator shown in Figure 5 to limit the maximum load current to be 700 mA. (*07 Marks*)

(d) Draw the circuit diagram of a basic op-amp shunt regulator and explain its operation. (*06 Marks*)

(e) Explain how a basic op-amp shunt regulator offers inherent short-circuit protection. (*04 Marks*)

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